

Chapter 67. Economics of Ventilator Care

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Economics of Ventilator Care: Introduction

One of the most urgent targets for improved efficiencies in inpatient hospital care is critical care services. Between 2000 and 2005, critical care medicine beds increased in the United States by 6.5% (from 88,252 to 93,955), and occupancy rates increased by 4.5%.¹ In that period, critical care costs per day increased by 30.4% (from \$2698 to \$3518). In 2005, critical care medicine accounted for 13.4% of hospital costs in the United States and consumed 0.66% of gross domestic product. Patients requiring mechanical ventilation are among the largest consumers of critical care resources, and hospitals often experience financial losses in providing care for them. As many as 2.8% of hospitalized patients in the United States received mechanical ventilation in 2005, representing 2.7 episodes of mechanical ventilation per 1000 population.² Estimated national costs were \$27 billion. This chapter reviews the economic implications of mechanical ventilation. Basic principles of health economics are reviewed to provide a framework for interpreting health economic analyses related to mechanical ventilation. Actual costs of mechanical ventilation are addressed, followed by a discussion of whether mechanical ventilation is cost-effective. Finally, strategies for cost containment are reviewed.

Basic Principles of Health Economics

The goal of *health economics* is to ascertain the highest level of efficiency in providing health care.^{3,4} A key assumption in this field is that health resources are a finite commodity. In such a system, a series of questions should be answered regarding any new or current medical intervention:

Is the intervention effective relative to other available therapies?

How much does it cost relative to other available therapies?

From whose perspective are the costs being considered?

How widely will the intervention be utilized?

Measured approaches to answering these questions allow health care systems to select medical therapies based upon evidence rather than assumptions, commercial marketing, or bias. Economic analysis has become a standard component of decision making for health systems in countries such as the United Kingdom or Australia, where health care policymaking is centralized on a national level. In countries such as the United States, delivery of health care is much less regulated, and many practitioners and most patients have unbounded access to any available therapies. Few physicians in the United States, however, are able to practice without significant awareness of the resource implications of their decision making. One goal of recent health care reform efforts in the United States is cost control, and it is highly likely that third-party payers will increase efforts to balance available services with more efficient delivery. The formation of Accountable Care Organizations, in which providers are incentivized to organize care delivery in a way that improves quality and outcomes and reduces costs, is mandated in the recent Affordable Care Act.⁵ Other delivery reform initiatives include pay-for-performance measures,⁶ expansion of medical homes, bundled payments, and value-based purchasing. Consequently, a basic understanding of how the efficiency of various health care practices is defined is becoming essential to the practicing clinician. The following section outlines some of the common definitions and methodologies employed in health economic analysis.

Costs

When considering the cost of therapies, the unit price of a drug or piece of equipment is often the basis of discussion in the clinical setting, especially if the analysis is being performed in the interest of the physician or hospital. Payers may have a broader view, especially if they are responsible for health care costs after hospital discharge. Patients and society view the cost of illness from an even larger perspective that includes a longer time horizon. Ideally, economic analyses determine the true costs that are accrued from the beginning of a patient's disease to the long-term outcome for the patient. These costs are categorized as direct medical costs, direct nonmedical costs, and indirect costs.^{3,4,7}

Direct Medical Costs

Direct medical costs are expenses for the provider of a service. These can include *variable costs* such as physician labor, drug costs, or use of diagnostic or therapeutic equipment.⁸ Variable costs are not generated unless the service is provided. An important subgroup is *direct variable costs*,

which are defined as variable costs excluding staff salary and equipment costs.⁹ Nursing labor and equipment are usually considered *fixed costs*, as nurse salaries must be paid regardless of bed occupancy, unless staff are laid off. Another form of fixed costs that is more removed from medical practice is *overhead costs*, which include general operations of the hospital or clinic such as utilities, leases, capital improvement, insurance, and administrative overhead or taxes.

Direct Nonmedical Costs

Direct nonmedical costs include costs incurred by the patient that are not related to care provided by the physician or hospital. These costs may include emergency transportation to the hospital, transportation and lodging for the patients' family, and domestic help and rehabilitative services after discharge. Although these costs are difficult to measure, they can have an important impact on the patient and their perception of the benefits of care.

Indirect Costs

Indirect costs include the overall financial burden of illness to the patient. This can include loss of wages and benefits as a consequence of missed work or loss of earnings and unpaid care provided by family members. This, too, is difficult to measure, but it has substantial impact on the well-being of the patient and family.^{10–13} Indirect costs for the patient can ultimately affect direct costs for a health system. If economic burdens on the patient decrease access to resources, affect medical adherence, and result in incomplete recovery, then recurrence of hospitalization or even critical illness can result.

Measuring Costs

One of the greatest challenges in health economic analysis is accurate measurement of relevant costs. Many hospitals have adopted sophisticated cost-accounting systems that assign a specific cost to each service based on hospital expenses.⁸ Expenses can be determined from either the hospital's acquisition costs or the service-specific relative-value units based upon the Center for Medicare and Medicaid Services Resource-Based Relative Value Scale. Overhead costs are assessed internally and assigned to each department.

Most hospitals increase the patient charge for many services above acquisition costs and expenses to help pay for services that are otherwise inadequately reimbursed. Because this practice can vary according to different payment plans, hospital *charges* are often an inappropriate surrogate for hospital *costs*.¹⁴ Large studies involving patients under a single-payer such as Medicare can adjust charges using a standard cost-to-charge ratio to gain acceptable estimates of costs, but this is less reliable with more heterogeneous groups of patients and payers.

As nursing time, monitoring costs, physician costs and certain laboratory and radiology ordering practices are linked to a patient's presence in an intensive care unit (ICU), ICU length of stay is considered by many to be a reliable and convenient surrogate for ICU costs. In one frequently cited model that is based on the relative costs per day of hospital admission, 4.5 units are assigned to the first day of each ICU stay, 2.5 units for each additional ICU day, and 1 unit for each non-ICU day after the first ICU discharge.¹⁵

For comparisons between institutions, length of stay can be confounded by differences in ICU admission criteria, nurse-to-patient ratios, presence of intermediate care or "stepdown" units, and transfers to other acute care facilities. To standardize resource use between different ICU settings, instruments such as the Therapeutic Intervention Scoring System (TISS) can be used.¹⁶ The TISS instrument assigns weights to seventy-six ICU interventions based upon the severity of illness associated with the need for each intervention. Intensity of ICU care can then be assessed by comparing the added weights or TISS scores between intervention groups, and costs can be standardized by applying a specific cost to each service.

Cost data for economic evaluations can come from multiple sources, but the most common approach is to collect cost data prospectively during clinical trials.^{17,18} Utilizing cost data from a single, large clinical trial has the advantage of uniform methodology, but it may not be generalizable if the trial involved highly selected patients managed in ways that are not standard to the typical clinical setting. When clinical trial data are not available, cost and outcome data can be derived from cohort studies. With this approach, sensitivity analyses to account for clinical variability and potential biases in the cohort studies are particularly important. A combined approach that begins with data from a large clinical trial and adds multiple scenarios that account for clinical variability is useful.

Types of Economic Analyses

Economic analyses function to provide more than just a description of the costs of a therapy. Their role is to compare costs between therapies relative to the benefits or efficacies of those therapies. Table 67-1 describes the four primary types of economic analyses.^{3,4,17} Cost-effectiveness analysis, the most useful approach to economic analysis, is described in more detail below.

Table 67-1: Types of Economic Analyses

Type	Comparisons	Result	Advantages/Disadvantages
Cost-Minimization	Costs of therapies with similar efficacies	Cost difference	Few therapies have similar efficacies for multiple outcomes
Cost-Benefit	Monetary value of costs and benefits of therapies	Cost/monetary benefit	Therapies for different conditions and outcomes can be compared, but it is difficult to place a monetary value on clinical outcomes such as death or quality of life
Cost-Effectiveness	Therapies with efficacies measured using similar clinical outcomes	Cost/clinical outcome (e.g. life-years saved)	Utilize clinically relevant outcomes. Analyses using different outcomes cannot be compared
Cost-Utility	Therapies for which mortality and quality of life are important outcomes	Cost/QALY	Different therapies with different efficacies can be compared; includes patient-centered outcomes; difficult to comprehend

Abbreviation: QALY, quality-adjusted life-years.

Cost-Effectiveness Analyses

Cost-effectiveness analyses use standard clinical measures such as life-years gained or quality of life units rather than monetary units to assess benefit. Outcomes are expressed as a ratio of cost to measure of benefit, for example, cost per life-year gained. Cost-effectiveness analyses are easy to understand for clinicians and have the advantage of not having to assign a monetary value to outcomes. Consequently, they are the most common type of economic evaluation performed. They can, however, only compare the costs of therapies that have a common clinical outcome.

Cost-Utility Analyses

An important and common subgroup of cost-effectiveness analysis is cost-utility analysis, which is performed when therapies are likely to have an impact on quality of life as well as mortality. The level of well-being for a given health state (*utility*, rated from 0 for worst health or death to 1 for best health) is multiplied by the amount of time spent in that health state. The resulting index is called a *quality-adjusted life-year* (QALY). For example, a therapy that results in a health state valued at 0.5 for 2 years would yield 1 QALY. Utilities can be measured by interviewing participants of a clinical trial, or predetermined values measured in similar groups of patients can be utilized. Results of a cost-utility analysis are expressed as cost per QALY. Results of selected cost-utility analyses related to mechanical ventilation and ICU care are listed in [Table 67-2](#).

Table 67-2: Cost-Effectiveness of Selected Mechanical Ventilation-Related Interventions

Intervention	Cost/QALY
Mechanical ventilation for respiratory failure related to pneumonia or ARDS ^{28,86}	\$29,000 (estimated 2-month survival > 70%) \$44,000 (estimated 2-month survival 51%–70%) \$110,000 (estimated 2-month survival ≤ 50%) \$32,000 (age < 65 years) \$46,000 (age > 75 years)
ICU care for patients with acute respiratory failure <i>without</i> chronic lung disease ²⁷	\$11,970 ^a
ICU care for patients with acute respiratory failure <i>with</i> chronic lung disease ²⁷	\$14,365 ^a
Mechanical ventilation to treat acute respiratory failure ⁸⁷	€1391 ^a
Prolonged mechanical ventilation (>21 days) after acute illness ³⁰	\$82,411 \$60,967 (estimated 1-year survival > 50%) \$101,787 (estimated 1-year survival ≤ 50%)
Lung-protective ventilation protocol in patients with acute lung injury ⁵²	\$11,690
ICU care for patients with acute renal failure ²⁷	\$30,625 ^a
Activated Protein C for severe sepsis ⁸⁸	\$27,400 (APACHE II ≥ 25) >\$100,000 (APACHE < 25)
Ultrasound screening for deep vein thrombosis in ventilated patients with femoral vein catheters ⁸⁹	\$12,793
Lung transplantation compared to standard care ⁹⁰	\$44,000 (assuming 10-year survival) \$204,000 (assuming 5-year survival)
In-hospital cardiopulmonary resuscitation	\$215,000

^aAnalysis did not include costs after hospital discharge.

Abbreviations: ARDS, acute respiratory distress syndrome; ICU, intensive care unit; QALY, quality-adjusted life-years.

The advantages of cost-effectiveness analyses are that costs and benefits of different therapies involving different types of patients can be compared, and outcomes beyond survival are factored in. The disadvantage is that many clinicians have difficulty understanding the complexity of the studies, and the models can involve numerous assumptions that vary across patient populations. Therefore, strict guidelines¹⁹ must be followed in conducting the studies to ensure transparency in how utilities are assigned, and rigorous testing in the form of sensitivity analyses should be performed to determine how variability in clinical factors could affect results of the models.

Cost-effectiveness analyses can aid decision making by providing data on how much therapies cost relative to the outcomes that are achieved, particularly if one therapy has greater benefits than the alternative, but the costs are greater as well. Cost-effectiveness analysis is not a cost-containment tool. Instead, it should be considered a method to improve value. The analyses may not always lead to decreased costs for a health care system, especially when underutilized services that increase cost but improve value are identified and encouraged.²⁰ In most Western health systems, an incremental cost per QALY of less than \$20,000 is strong evidence for adoption of a therapy. Incremental costs per QALY of between \$50,000 and \$100,000 provide moderate evidence for adoption. If the incremental cost per QALY is greater than \$100,000, the therapy may not be considered cost-effective, although more recent recommendations raise that boundary to a range of \$183,000 to \$264,000.²¹ These boundaries of cost-effectiveness

are somewhat arbitrary however. Ultimate decision making must take into account other factors such as seriousness of the health condition, availability of alternatives, and number of patients who would receive the therapy (total budgetary impact).^{3,22}

Costs of Mechanical Ventilation in the Intensive Care Unit

The average total cost of hospitalization for a critically ill patient ranges from \$14,135 to \$32,253, depending on the study methods and patient population.^{23,24} This is nearly three times the cost of a hospitalized patient managed on the medical or surgical floor. Two-thirds of the costs associated with critically ill patients are accrued during their stay in the ICU. For those whose length of stay in the ICU is more than 5 days, as much as 80% of their hospital costs are accrued in the ICU. Hospital costs for ICU patients who require mechanical ventilation is significantly higher than for nonventilated patients in the ICU (e.g., \$47,158 vs. \$23,707 in one study).²³ It is important, however, to understand that the difference in cost between ventilated and nonventilated patients is a factor of higher illness severity rather than the cost of providing mechanical ventilation.²⁵

Maintaining a mechanical ventilator accounts for less than 5% of direct ICU costs for a ventilated patient.⁹ A ventilator includes a one-time cost of \$20,000 to \$45,000 for the hospital plus nominal maintenance charges, and it is used for a number of years. A respiratory therapists' time ranges from \$70 to \$130 per day depending on local salary and staffing levels, and administrative costs are a smaller factor. Rather than the mechanical ventilator itself, the major contributor to variable costs for a ventilated patient is the nursing effort assigned to the patient (Table 67-3).^{26,27} The nurse-to-patient ratio for acutely ill, mechanically ventilated patients is usually 1:1 or 1:2.

Table 67-3: Costs of Managing Intensive Care Unit Patients by Human and Capital Components

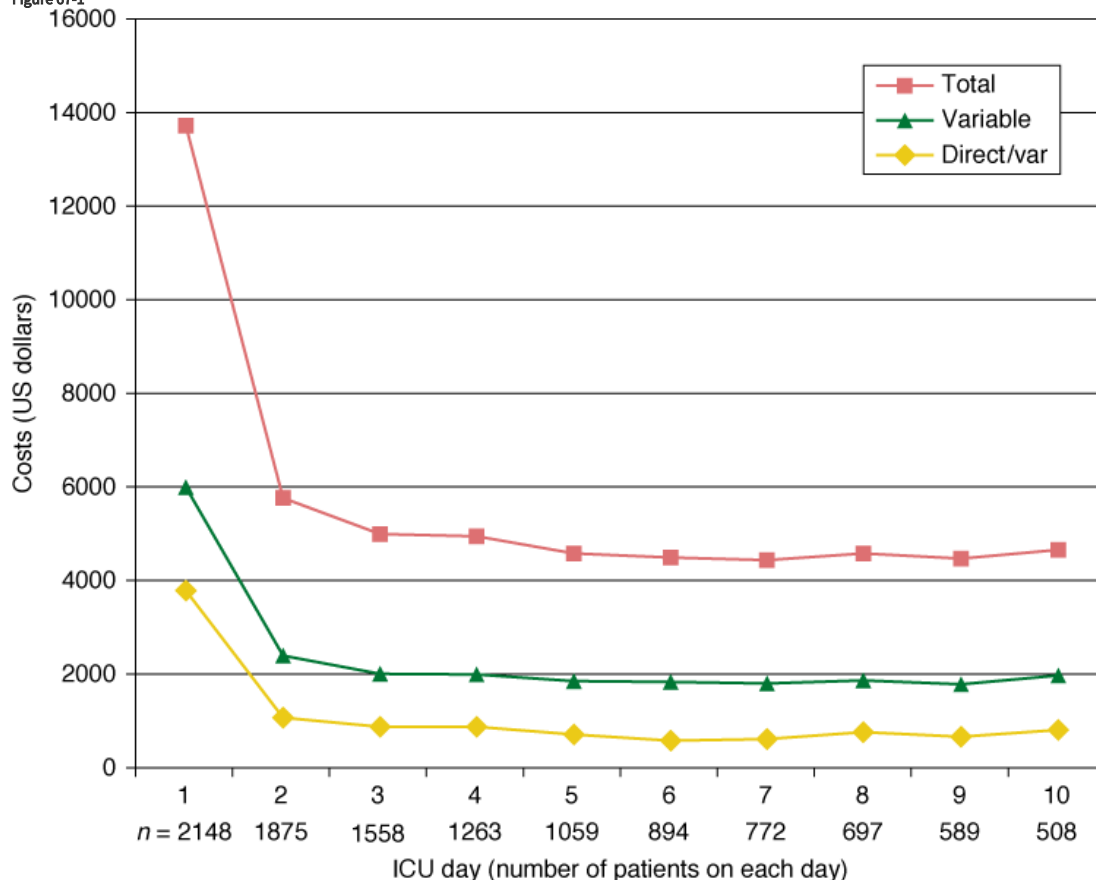
Component	% Component Cost	% Total Cost
Human	–	63.8
Nursing	64.6	41.2
Medical	15.1	9.6
Professional	14.3	9.1
Support	6.1	3.9
Supplies	–	11.7
Laboratory	–	11.4
Chemistry	62.0	7.1
Hematology	23.5	3.3
Microbiology	9.9	1.1
Other	4.6	0.5
Medication	–	7.4
Diagnostic imaging	–	4.5
Capital equipment	–	1.2

Source: Used, with permission, from Noseworthy et al.²⁶

One study assessed the fraction of total costs attributable to variable and direct-variable costs for mechanically ventilated patients at an urban teaching center.⁹ The average total cost for each patient was \$69,472. Costs were highest during the first 2 days of intensive care and decreased

significantly thereafter (Fig. 67-1). Only 18.4% were direct-variable costs (costs not attributable to overhead, staff, or equipment). Direct-variable costs were highest for the blood bank (44%) and pharmacy (48%), and lower for radiology (8%) and respiratory care (3%).

Figure 67-1



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Mean costs by ICU day for patients receiving mechanical ventilation. Mean total (*Total*), variable (*Variable*), and direct-variable (*Direct/var*) costs by day for each ICU admission. (Used, with permission, from Kahn et al.⁹)

Is Mechanical Ventilation Cost-Effective?

Thorough economic analyses of mechanical ventilation have been limited because of difficulties in applying adequate study methods in this complex patient population. One high-quality study, however, addressed the question of cost-effectiveness of mechanical ventilation in severely ill patients who required mechanical ventilation for pneumonia or acute respiratory distress syndrome (ARDS).²⁸ In the analysis, patients were stratified based upon likelihood of 2-month survival (Table 67-4). The incremental cost per QALY of providing mechanical ventilation to these patients was \$29,000 for low-risk patients (>70% estimated survival), \$44,000 for medium-risk patients (51% to 70% estimated survival), and \$110,000 for high-risk patients (≤50% estimated survival). Sensitivity analyses that increased mortality and costs to twice the baseline estimates resulted in incremental costs per QALY that were still less than \$80,000 for low-risk and medium-risk patients. In another cost-utility analysis from France,²⁷ cost-utility ratios for ICU care of patients with acute respiratory failure were estimated to be \$11,970 for patients without chronic lung disease and \$14,365 for patients with chronic lung disease (see Table 67-2). This analysis did not include costs of care following discharge, so it is relevant only from the hospital's perspective.

Table 67-4: Costs of Mechanical Ventilation for Patients with Acute Respiratory Failure Secondary to Pneumonia or Acute Respiratory Distress Syndrome

	>70% Estimated 2-Month Survival	51% to 70% Estimated 2-Month Survival	≤50% Estimated 2-Month Survival
Hospital costs	\$59,096 ± \$64,336	\$70,130 ± \$85,300	\$59,310 ± \$54,590
Physician costs	\$5,034 ± \$6,705	\$6,162 ± \$5,264	\$6,474 ± \$7,426
Costs through the year after discharge	\$22,037 ± \$44,847	\$18,772 ± \$42,253	\$11,994 ± \$34,475
Annual costs after first year	\$18,265 ± \$48,078	\$13,053 ± \$26,519	\$28,102 ± \$86,998
One-year survival	0.62	0.39	0.21
Cost per QALY	\$29,000	\$44,000	\$110,000

Patients were enrolled in the SUPPORT (Study to Understand Prognoses and Preferences for Outcomes and Risks of Treatments) trial. Index hospital costs were derived from hospital charges using Medicare cost-to-charge ratios. Physician costs and costs of hospitalization after the index hospital discharge were estimated from Medicare financial data. Two-month survival estimates were determined at the time of diagnosis of acute respiratory failure.

Abbreviation: QALY, quality-adjusted life-year.

Source: Used, with permission, from Hamel et al.²⁸

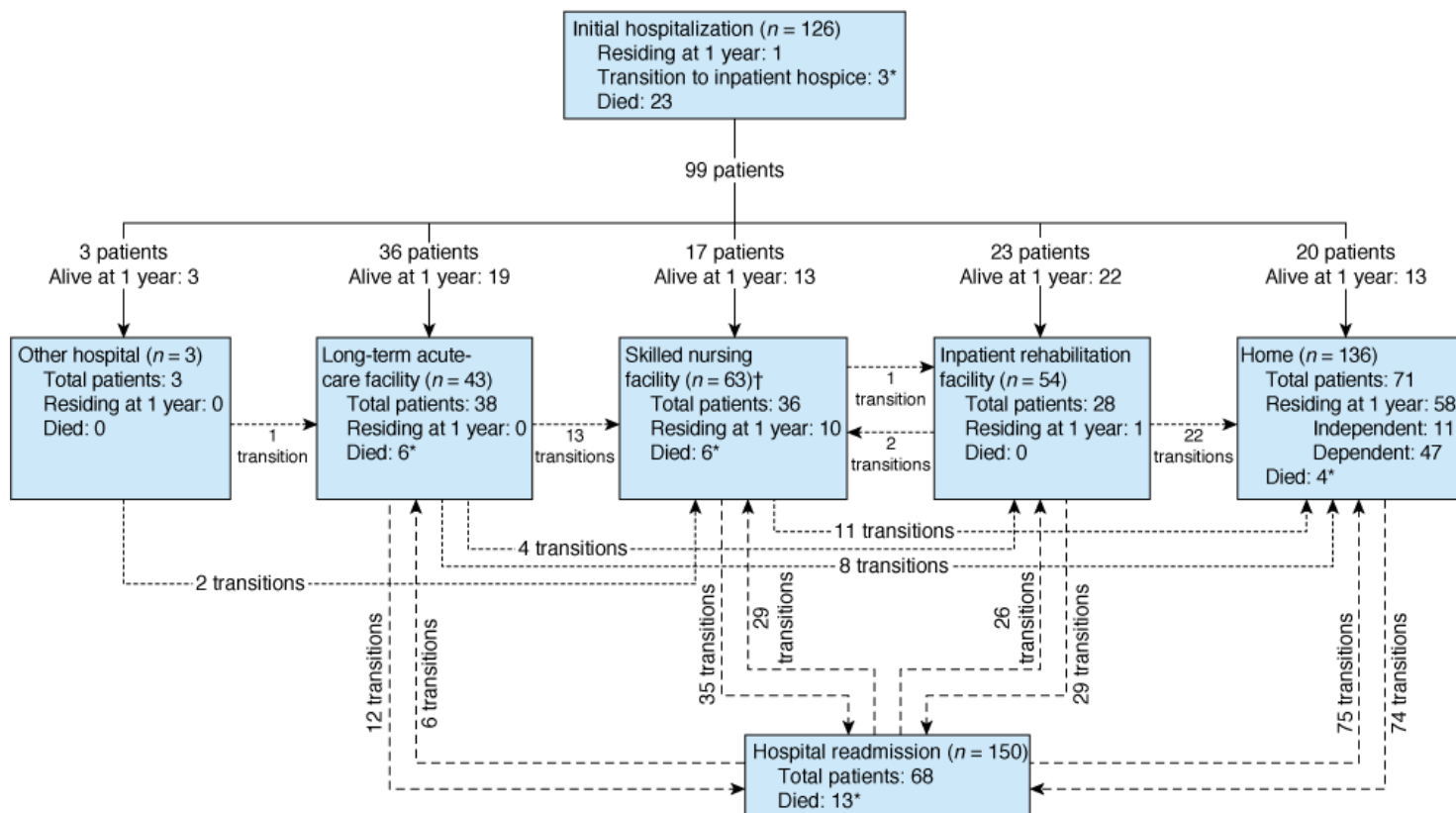
These data suggest that mechanical ventilation for patients with acute respiratory failure as a consequence of pneumonia or ARDS meets standard criteria for cost-effectiveness for all but the highest-risk patients. It may be reasonable to assume that mechanical ventilation for patients with conditions that afford better prognoses, such as asthma, chronic obstructive pulmonary disease (COPD), or routine postoperative patients, will be similarly cost-effective, and mechanical ventilation for patients with worse prognoses, such as pulmonary malignancies or hematologic malignancies, may be less cost-effective. Of course, it is rather difficult to apply these principles in bedside practice. When resources are available, mechanical ventilation should be utilized according to how the expected benefit matches the patient's wishes for invasive care.

Costs of Prolonged Mechanical Ventilation

Patients who require more than 7 days of mechanical ventilation represent only 10% of ICU patients, but they consume up to 40% of ICU resources.²⁹ More than 20% of those resources are consumed after the seventh day of ventilation. In 2005, patients requiring prolonged mechanical ventilation ranked third in summative inpatient charges by diagnostic group and first in diagnostic charges per patient.³⁰ The number of patients in the United States requiring at least 7 days of mechanical ventilation has been increasing steadily since 2000, and by 2020, the number is projected to double.³¹ This is projected to require an absolute increase of over 2.1 million mechanical ventilator days, 3.2 million ICU days, and 6.5 million hospital days over year 2000, at a total inflation-adjusted cost over \$64 billion.³²

Health care costs associated with prolonged ventilation remain high even after patients are liberated from mechanical ventilation and discharged from the hospital. Readmissions occur in up to 40% of patients within a year, and total health care costs that accrue for patients during the year after discharge are almost as high as their index hospital costs because much of their time is spent in institutional care such as nursing homes or rehabilitation facilities.³³ In one single-center cohort study, prolonged ventilation patients who survived their acute hospitalization experienced a median of four transfers between different care centers, and 74% of days alive were spent in institutional care (Fig. 67-2).³⁴

Figure 67-2



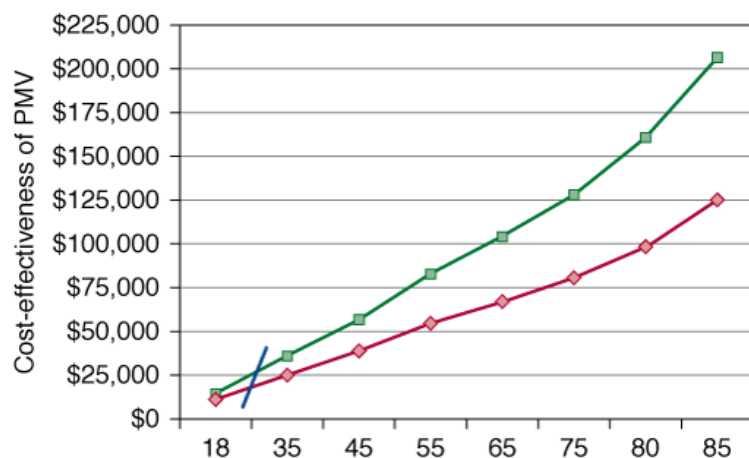
Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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Trajectories of care over the first year after discharge for a cohort of patients requiring prolonged mechanical ventilation in a U.S. tertiary care hospital. *Arrows* between locations indicate both the direction of patient transitions and the total number of patients transferred between locations over 1 year. *Solid lines* represent initial transitions between the hospital and other locations. *Dashed lines* represent subsequent hospital readmissions and discharges involving postdischarge care locations. *Dotted lines* represent transitions among postdischarge care locations, including home. Each box summarizes the total numbers of both readmissions and patients admitted, as well as how many patients remained or died in each location of care at 1 year. *, Seven transitions to inpatient hospice and death not shown (three from the acute hospitalization and one each from home, long-term acute care facility, skilled nursing facility, and hospital readmission); †, one transition from skilled nursing facility to skilled nursing facility not shown. (Used, with permission, from Unroe et al.³⁴)

One-year survival of patients requiring prolonged ventilation ranges from 23% to 50%.^{35,36} Considering their poor long-term outcomes and very high costs, the question inevitably arises as to whether or not prolonged care for these patients is cost-effective. In one economic analysis, investigators measured the cost-effectiveness of mechanical ventilation beyond 21 days compared to withdrawal of life-sustaining therapies as a hypothetical comparison.³⁰ They determined that providing prolonged mechanical ventilation cost \$82,411 per QALY. Cost-effectiveness ratios were most sensitive to variation in age, hospital costs, and probability of readmission (Fig. 67-3).

Figure 67-3



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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Incremental costs per life-year (red line) and incremental costs per quality-adjusted life-year (green line) of prolonged mechanical ventilation (PMV).

(Used, with permission, from Cox et al.³⁰)

Controlling Costs of Mechanical Ventilation

From the perspective of patients and society, mechanical ventilation results in significant and sometimes catastrophic charges, especially when provided in the ICU of an acute hospital. From the hospital perspective, more than half of the patients managed in the ICU generate costs that are greater than the average payment in the Medicare Diagnostic Related Group (DRG) system, leading to financial losses for the hospital.²⁴

To control costs associated with mechanical ventilation, a patient's time on the ventilator and in the ICU should be limited as much as safely possible. Strategies to reduce duration of mechanical ventilation include prevention of intubation, efficient management of the ventilated patient, and withholding or withdrawal of mechanical ventilation in terminally ill patients (Table 67-5).²⁵ For the hospital, savings can be achieved by transfer of the ventilated patient to lower-intensity sites of care. It should be noted that none of these strategies will significantly reduce overall costs if they merely shift costs from one cost center to another (e.g., from an ICU to a weaning facility).⁸ Furthermore, one less day of mechanical ventilation and ICU care does not amount to savings relative to the average daily cost of ICU care, but rather to the last day of ICU care, which is much lower than the average daily cost.⁹ And, finally, most of the costs associated with a day of mechanical ventilation are fixed costs rather than variable costs or direct-variable costs (see Fig. 67-1). Therefore, reducing time on a ventilator only lowers costs significantly if it leads to fewer days in the hospital and reductions can be sustained across enough patients over enough time to allow for closure of ICU beds and hospital beds.^{8,37} With these caveats in mind, some strategies to reduce costs of care for the ventilated patient are discussed below.

Table 67-5: Cost-Containment Strategies for Mechanical Ventilation

Accepted strategies

Prevention of intubation
 Rapid response teams
 Noninvasive mechanical ventilation
 Efficient management practices
 Daily screening and performance of spontaneous breathing trials
 Daily interruption of sedative infusions
 Use of short-acting sedatives
 Early mobilization
 Prevention of ventilator-associated pneumonia
 Prevention of intravenous catheter-related infections
 Intensive care unit (ICU) organization
 Critical care trained medical director
 Closed ICU format with full-time intensivist coverage
 Nurse-driven sedation protocols
 Nurse-driven or therapist-driven weaning protocols
 Flexible nurse staffing
 Continuous medical education
 Continuous quality improvement
 Telemedicine
 Transfer to lower-intensity sites of care
 In-house respiratory care units
 Long-term acute care hospitals or regional weaning centers
 Acute rehabilitation hospitals with mechanical ventilation capabilities
 Skilled nursing facilities with mechanical ventilation capabilities
 Home mechanical ventilation

Unproven strategies

Withholding or withdrawing mechanical ventilation from terminally ill patients
 Advanced directives are uncommon and do not influence practice
 Very poor prognoses are often not evident until well into the ICU course
 Cost savings are reduced if the patients require continued hospitalization for palliative care

Prevention of Intubation

One effective means of preventing mechanical ventilation in the ICU is to intervene with appropriate resuscitative efforts before respiratory failure occurs. Early recognition and intervention for conditions that will predispose patients to respiratory failure such as sepsis, cardiogenic pulmonary edema, or acute bronchospasm may reverse processes enough to avoid intubation. Rapid response teams can facilitate early interventions, but some models require significant additional staff and may not improve hospital survival.^{38–40} Protocols for early goal directed therapy in septic shock may reduce ICU resource requirements,⁴¹ but confirmation of early findings await results of ongoing clinical trials. Perioperative clinical management pathways have benefitted from fast-track anesthesia, and enhanced care in postanesthesia care units have greatly reduced the need for ICU care in cardiac, thoracic, and head-and-neck surgical patients.^{42–45}

Noninvasive Mechanical Ventilation

Noninvasive ventilation for acute respiratory failure secondary to severe COPD exacerbations decreases the likelihood that a patient will need intubation.^{46,47} This is associated with significantly lower hospital mortality and decreased ICU and hospital length of stay. Noninvasive ventilation has also been shown to decrease likelihood of intubation in patients with congestive heart failure without shock or ischemia, patients with community acquired pneumonia if they have underlying COPD, patients with acute-on-chronic respiratory failure associated with obstructive sleep apnea, and in immunocompromised patients who have early ARDS.⁴⁸

Avoiding intubation does not eliminate all of the costs associated with mechanical ventilation. Noninvasive ventilation in the setting of acute respiratory failure still requires a significant investment in respiratory therapist time in addition to equipment and administrative costs. More importantly, if the patient requires monitoring in an ICU, the actual cost savings will be reduced.⁴⁹ Many hospitals manage patients with acute respiratory failure requiring noninvasive ventilation on a stepdown unit or hospital floor that has lower nurse-to-patient ratios than in an ICU. Although achieving some cost savings, this approach should be implemented only if well-trained staff and sufficient monitoring are available to be able to respond to patients who fail noninvasive ventilation. Utilization of noninvasive ventilation in patients with less-severe COPD exacerbations would result in financial losses because it requires a higher level of monitoring than standard therapy and does not improve outcomes.⁵⁰

Efficient Management Practices

Because decreasing time on the ventilator and time in the ICU is the mainstay of cost reduction for mechanical ventilation, recent clinical trials have focused on these issues as important outcomes. Use of normal tidal volumes (6 mL/kg predicted body weight) in patients with acute lung injury or ARDS rather than tidal volumes of 12 mL/kg of predicted body weight results in fewer ventilator days in addition to lower mortality.⁵¹ As a result, it is a proven cost-effective therapy.⁵² Daily screening for readiness to wean and initiation of spontaneous breathing trials does reduce ventilator days by 1.5 days compared to usual care.⁵³ Similarly, daily awakening of patients receiving continuous infusions of sedatives can decrease ventilator days by 2.4 days compared to usual care.⁵⁴ Use of a short-acting sedative rather than long-acting sedatives can reduce ventilator days as well, in part by reducing delirium.^{55,56} These practices by themselves have a negligible impact on mortality, and total hospital costs are not dramatically impacted by a few less days of ICU care. But when these practices are applied in combination, long-term survival of patients can be significantly improved, indicating clear benefit.⁵⁷ Early mobility protocols are an extension of this concept of animating the ventilated patient to enhance cognitive and physical function.^{54,58,59} Because some of these practices have not been standard care at many medical centers, clinicians have been slow to adopt them for their daily routine. Practice guidelines and nurse-driven or therapist-driven protocols have been very useful in these settings to change practice.^{60–63} A meta-analysis evaluating the benefit of instituting formal weaning protocols indicated that they can result in decreased duration of mechanical ventilation and ICU stay, especially in surgical units, but mortality and hospital length of stay are not greatly affected.⁶⁴

Individual practices that decrease time on the ventilator may decrease the risk of ventilator-associated pneumonia. Ventilator-associated pneumonia can occur in up to 30% of ventilated patients, depending on the type of ICU. It is associated with longer time on the ventilator and longer ICU stays.⁶⁵ Ventilator-associated pneumonia may also result in higher patient mortality, although the attributable mortality of ventilator-associated pneumonia remains uncertain.⁶⁶ Therefore, prevention of ventilator-associated pneumonia is likely to be important for minimizing costs.⁶⁷ Early tracheostomy is another practice that is associated with fewer ventilator days and fewer days in the ICU. Most studies, however, do not demonstrate a mortality benefit, and not all studies result in fewer hospital days.^{68–70} None of the clinical trials assess the additional costs associated with tracheostomy placement and care, or how discharge disposition is affected (e.g., discharge to a nursing home rather than home because of tracheostomy care needs). Early tracheostomy remains an ongoing area of clinical research, with the largest challenge being accurate identification of patients who are likely to proceed to prolonged ventilation so that unnecessary tracheostomy can be avoided.

Intensive Care Unit Organization and Staffing

A dedicated critical care-trained medical director is essential to maintaining an efficient ICU, and a closed ICU organization with intensivists managing all aspects of patient care is the ideal. The rationale for maintaining high-intensity ICU staffing (mandatory intensivist consultation or a closed ICU setting) is supported by multiple studies in the literature, although none of them are randomized trials. A systematic review of studies comparing high-intensity staffing to settings with no intensivists or only elective intensivist consultation revealed a pooled estimate of relative risk for hospital mortality of 0.71 (95% confidence interval, 0.62 to 0.82).⁷¹ Although some studies noted greater use of ICU interventions, such as mechanical ventilation or pulmonary artery catheters, in the closed ICUs, high-intensity staffing reduced ICU length of stay in fourteen of eighteen studies, including the two studies using case-mix adjustment. Importantly, the degree of illness severity increased for many of the ICUs when they adopted high-intensity staffing, suggesting that dedicated intensivists achieve more appropriate use of ICU resources. Although the closed ICU model is the norm in Europe and Australia, only 30% of ICUs in the United States have adopted such forms of organization.⁷² This proportion may increase as quality initiatives by various health consortiums and industry groups have identified 24-hour intensivist coverage for ICU patients as an important priority.

Although a closed ICU model is advocated for better outcomes, there are limited data to clarify the best ratio of intensivists to patients. Furthermore, there will be an insufficient number of available intensivists in the United States to staff all ICUs in a closed model for the foreseeable future.⁷² Therefore, another relevant question is what type of clinicians, besides intensivists, provide the most efficient care. As a result of decreased availability of resident trainees, a number of models involving hospitalists, nurse practitioners, and physician assistants are being evaluated.^{73,74}

Another approach is to utilize evolving technologies in telemedicine. This expensive technology can be associated with improved hospital mortality and decreased ICU stay if there is broad physician acceptance, and if the intervention is focused on adherence to evidence-based practices and enforcement of daily patient goals.^{75,76}

Withholding or Withdrawing Mechanical Ventilation in Terminally Ill Patients

Up to 12% of health care expenditures and 27% of Medicare expenditures are spent on patients during the last year of their lives.^{77,78} For Medicare patients who die, 40% of their Medicare expenses are generated during the last month of their life. Seventy percent of their total expenses are spent on inpatient care. Dying patients often endure a period of critical illness before their death. Limiting ICU care for patients at the end of their life might seem like an obvious target for reducing ICU resource consumption and limiting health care costs, but realizing true cost savings in practice has been very difficult to accomplish for a number of reasons.^{37,79}

Advance directives are relatively uncommon and often countermanded in the setting of acute respiratory failure if palliative measures are not already in place. The SUPPORT (Study to Understand Prognoses and Preferences for Outcomes and Risks of Treatments) trial⁸⁰ and other studies have demonstrated that interventions to make physicians more aware of patient prognoses and preferences for end-of-life care do not alter outcomes or result in reductions in resource use. Once patients have been intubated, early withdrawal of support for those with the worst prognoses could potentially result in cost savings, but only if it resulted in death in a much shorter period of time than would have occurred with continued care. Prolonged palliative care in an inpatient setting would significantly reduce any incremental cost savings.

One study of an intervention to improve communication of prognosis and goals of care highlights the difficulties of achieving true cost savings from interventions that reduce ICU length of stay. In a before-and-after study design,^{81,82} investigators found that implementation of regular, structured family meetings emphasizing clinical milestones by the ICU team reduced median ICU length of stay from 4 to 3 days. ICU mortality rates also decreased, which the authors attributed to earlier triage of severely ill patients who were likely to die, with subsequent admission of less severely ill patients. So in effect, more efficient use of ICU beds in the care of severely ill patients did not result in any change in fixed ICU costs, because less-severely ill patients who were previously managed on the general wards were admitted to the ICU as more resources became available.

Transfer to Lower-Intensity Sites of Care

For critically ill patients, the highest intensity of care usually occurs within the first few days of ventilation. As gas exchange and hemodynamics improve, the level of nursing required to manage them decreases. As long as they remain in an ICU, however, the associated nursing commitment and costs remain substantial. When the lengths of stay extend beyond approximately 14 to 21 days, hospitals begin to assume considerable financial losses. Therefore, hospitals are quite motivated to find less-intensive sites of care for these patients.

Options for lower-intensity sites of care for patients requiring prolonged ventilation include (a) stepdown units or dedicated respiratory care units within the acute hospital, (b) transfer to separate acute care hospitals dedicated to weaning patients from mechanical ventilation such as regional weaning centers, long-term acute care hospitals (LTACs) or specialized rehabilitation hospitals, (c) transfer to skilled nursing facilities that offer ventilator care, or (d) discharge to home with home ventilation.⁸³ When ventilated patients are transferred from the acute ICU to stepdown units, some variable cost-savings are achieved primarily because of lower nurse to patient ratios. Prolonged weaning of weak patients demands a high degree of therapist monitoring, and aggressive physical rehabilitation during weaning is felt to be important for good outcomes. Therefore, in well-managed weaning units, savings in nursing costs are balanced by additional expenses for respiratory therapists and physical and occupational therapists. Custodial care of ventilated patients in skilled nursing facilities that does not include active weaning or rehabilitation is less expensive on a daily basis, but complete failure to wean patients results in higher long-term costs.

Hospitals realize the largest cost savings and open the most ICU beds by transferring ventilated patients to freestanding weaning hospitals, such as LTACs. These financial and resource incentives have driven considerable growth in the largely for-profit LTAC industry in the United States, resulting in an 8.8% increase in the number of LTAC hospitals per year between 1997 and 2006.⁸⁴ The number of patients transferred to LTACs for prolonged ventilation increased from 10,389 in the period between 1997 and 2000 to 30,877 between 2004 and 2006. It should be emphasized that transferring a ventilated patient to a freestanding weaning facility will result in cost savings for the hospital, but not necessarily for the health care system. Recent analyses sponsored by the Center for Medicare and Medicaid services have indicated that for patients transferred to LTACs, costs for the total episode of care (hospital and LTAC) were significantly higher than for similar patients whose total episode of care was managed completely at the acute care hospital. Whether this is secondary to additional costs and complications of transfer, less efficient management at the LTACs, or selection bias in the analysis, it certainly suggests that transferring patients to LTACs may ultimately result in cost shifting rather than overall cost savings.

The Future

Continued advances in technology will increase the gap between what critical care medicine can do and what is economically feasible to do. The formal application of health economics to guide decision making in the delivery or organization of critical care is in its early stages. Economic analysis will become more influential as providers become more familiar with its benefits and its limitations. Investigators can improve upon limitations by applying better methodology, especially relating to measurement of costs across various levels of care. Increasing sophistication of clinical information systems and inclusion of economic components in the design of clinical trials will provide higher quality data.^{17,18} Investigators are continuing to refine quality-of-life assessments, and better understanding of long-term outcomes of critically ill patients is an area of growing research interest. More research into the behavioral aspects of medicine including social and professional expectations for distribution of resources will provide guidance to societies and health plans on how to utilize the data from economic analyses.⁸⁵

Economic pressures and financial constraints will continue to have a beneficial effect on efficiency of care in the ICU by reducing waste and redundant services. ICU directors, however, need to exercise caution in the persistent drive for efficiency. Overly aggressive measures to discharge patients from the ICU and maintain high occupancy of only severely ill patients can worsen outcomes. For example, discharges from the ICU at night and ICU admissions during periods of peak occupancy and high nursing workload are associated with higher hospital mortality. From the perspective of society and health care systems, savings generated from reductions or restrictions in critical care services are justified only if those savings are invested in more effective services elsewhere in the system.

Summary and Conclusions

The high costs of critical care services will always make this area of medicine a target for cost containment. Patients requiring mechanical ventilation consume a disproportionate share of hospital resources, especially those requiring prolonged ventilation. Despite the costs, mechanical ventilation for patients with acute respiratory failure meets current standards for cost-effectiveness except for patients with the highest likelihood of short-term mortality. Strategies to reduce hospital costs are directed toward decreasing the duration of mechanical ventilation. These strategies include prevention of intubation by rapid response to illness and noninvasive ventilation when indicated, efficient evidence-based management practices, and optimal ICU organization including continuous quality improvement. Economic analyses will assume a greater role in decision making related to mechanical ventilation as investigators include cost and quality-of-life components in the design of clinical trials and as providers gain a better understanding of how to utilize the data.

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